

rocket fuel. Studying the atmospheric dust size and morphology will provide data on whether surface systems can function in such harsh conditions. More data about the planet's atmospheric model will also be acquired with this study. This will help us observe the effects of the external conditions on human health as well. Few of the last objectives would be to measure data on aerothermal conditions, thermal protection system, and aerodynamic performance on entry vehicles. These data sets are necessary since they will enable scientists to design spacecraft for a safe entry and descent into the Martian atmosphere.

IMPROVING CURIOSITY

The new rover will be based on the fundamental design of Curiosity. This ensures lower mission costs and even lower risks. However, several upgrades have been made to ensure it delivers the required objectives. Curiosity had an arm which was used to collect rock and soil samples from the surface and now Mars 2020 adds a drill. Having a drill makes it easier to penetrate Martian rocks and soil to collect samples for analysis. These samples will be stored in tubes on the surface through a process called "depot caching". Essentially, the rover will directly collect or drill down rocks and store these samples into tubes. The tubes will be sealed and placed on storage racks right on the surface of Mars, waiting to be picked up in a future retrieval mission. Another improvement applied to the Curiosity design is an upgraded wheel design which will probably favour easier traversal over different terrains of the Martian surface.

Mars 2020 will employ the same method applied by Curiosity for its entry, descent and landing (EDL) on Mars. The spacecraft will use a three-stage system including a parachute, descent vehicle, and "sky crane maneuver". The third final stage is an approach to ensure the spacecraft stabilises over the surface before the rover is lowered using tethers. In this way, a more precise landing is achieved. Mars 2020 uses a new system called Terrain-Relative Navigation (TRN) so that the rover detects unfavourable and hazardous terrain below, and looks for a safer landing spot.



(Credits: NASA/JPL-Caltech)

"Sky crane maneuver" in the landing state

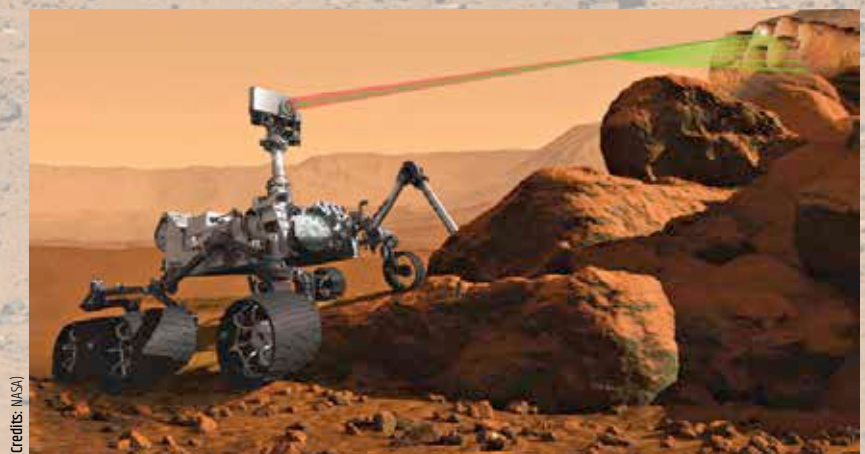
NEW INSTRUMENTS

To meet the objectives of the mission, several instruments were proposed to be installed on the Mars 2020. Out of all of them, seven instruments have been

selected that will carry out the necessary research.

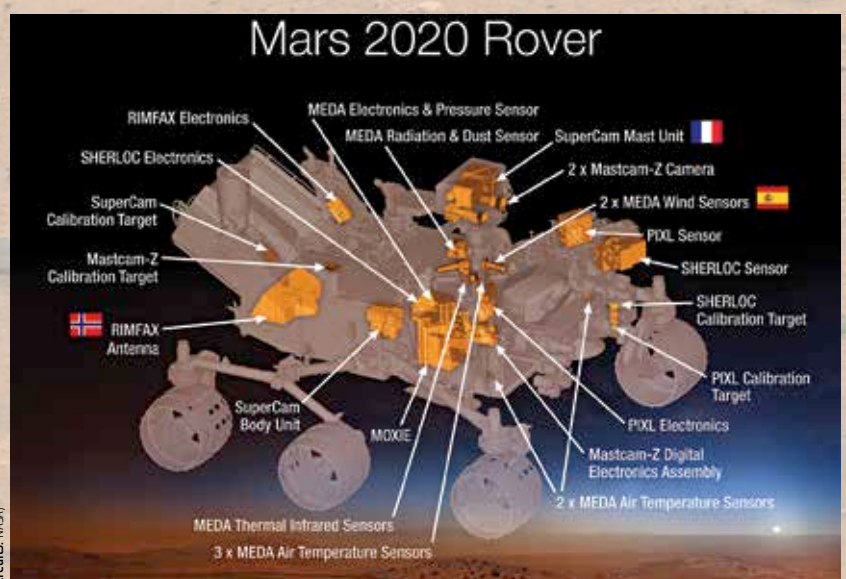
Images from the Martian surface are always exciting to look at which will be made possible through the Mastcam-Z. Two units of this advanced camera system will allow the rover to take panoramic and stereoscopic images including the functionality to zoom. Not only pictures, the camera system will be able to measure the mineralogy of the rocks and soil on the surface, and assist with rover operations. SuperCam is another instrument that has imaging, chemical composition analysis and mineralogy functionalities. Hence, it will be able to determine the chemical composition of rocks and detect the presence of organic compounds without even touching them.

Among the two spectrometer instruments on the rover, the PIXL or Planetary



(Credits: NASA)

Artist's rendering of the SuperCam instrument on board the rover



(Credits: NASA)

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